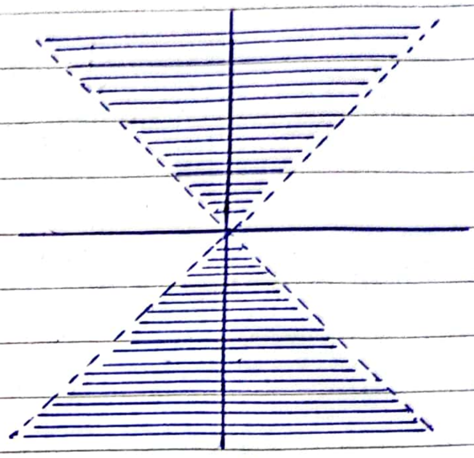
Q# 65

$$f(x,y) = \sum_{n=0}^{\infty} \left(\frac{x}{y}\right)^n \qquad (1,2)$$

By applying formula of geometric series;

where $|r| < 1$



$$f(x,y) = \frac{1}{1 - \frac{x}{y}}$$

for Domain $\left|\frac{x}{y}\right| < 1$

So,

Domain: $|x| < |y|$; all Points (x,y) satisfying the inequality.

$$f(x,y) = \frac{1}{1 - \frac{x}{y}}$$

$$f(x,y) = \frac{1}{\frac{y-x}{y}}$$

$$f(x,y) = \frac{y}{y-x}$$

as $f(x,y) = c$,, So

$$c = \frac{y}{y-x}$$

at $(1,2)$

$$c = \frac{2}{2-1}$$

$$\boxed{c = 2}$$

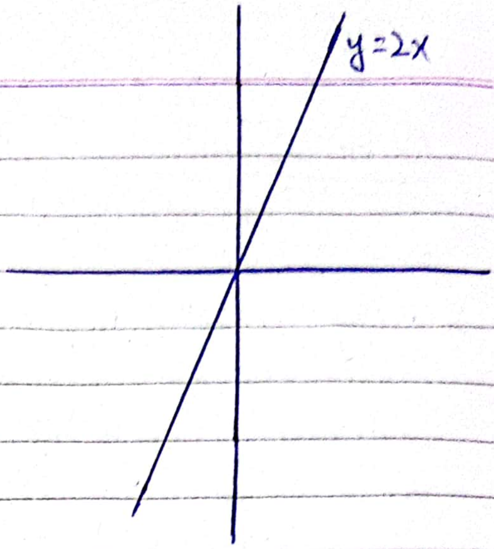
So,

$$2 = \frac{y}{y-x}$$

$$2y - 2x = y$$

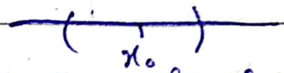
$$2y - y = 2x$$

$$\boxed{y = 2x}$$



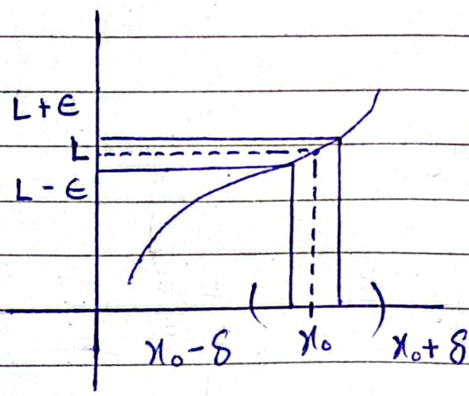
Ex#14.2 Limits & Continuity in Higher Dimensions

$$\lim_{x \to x_0} f(x) \longrightarrow L$$

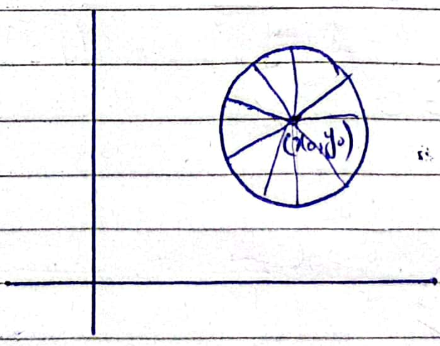


neighborhood

$$\boxed{|f(x) - L| < \epsilon}$$



Neighborhood of x_0 are open intervals.



Neighborhood of (x_0, y_0) are open discs.

$$\sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$$

Limit of function of two Variables:

We say that the function $f(x,y)$ approaches the limit L as (x,y) approaches (x_0, y_0) and write

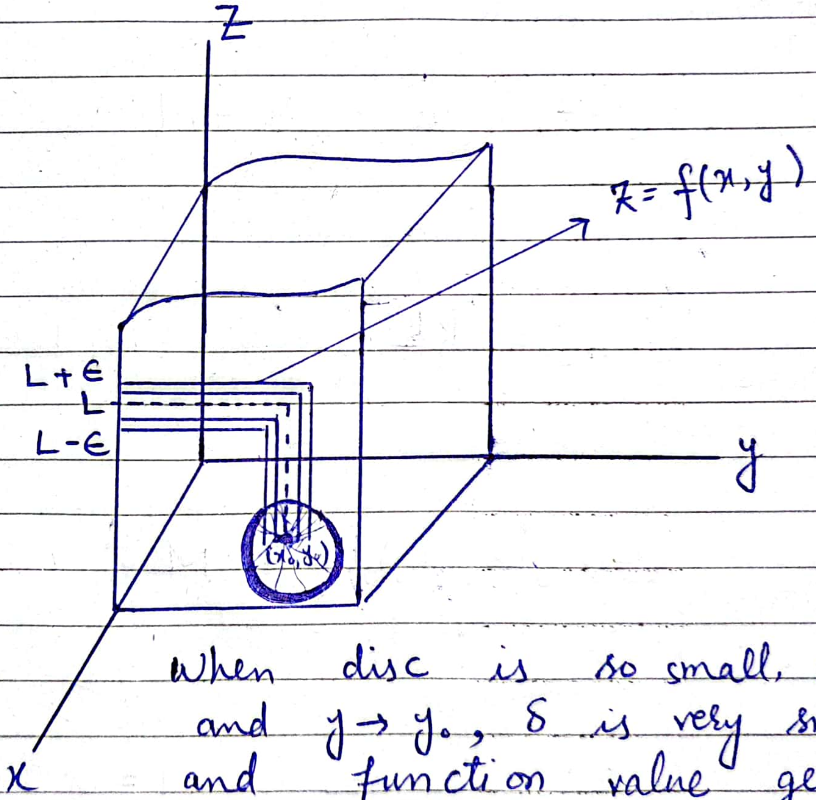
$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) \to L$$

If, for every chosen $\epsilon > 0$ there exist a corresponding $\delta > 0$ such that for all (x,y) in the domain of $f(x,y)$

$$|f(x,y) - L| < \epsilon$$

When

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$$



When disc is so small, $x \to x_0$ and $y \to y_0$, δ is very small, and function value gets closer to L, for a chosen ϵ.

Theorem: Properties of Limit of functions of two variables.

The following rules hold if L, M, K are real numbers and

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) \to L \quad , \quad \lim_{(x,y)\to(x_0,y_0)} g(x,y) \longrightarrow M$$

1. **Sum Rule:**

$$\lim_{(x,y)\to(x_0,y_0)} [f(x,y) + g(x,y)] = \lim_{(x,y)\to(x_0,y_0)} f(x,y) + \lim_{(x,y)\to(x_0,y_0)} g(x,y)$$

$$= L + M$$

2. **Difference Rule:**

$$\lim_{(x,y)\to(x_0,y_0)} [f(x,y) - g(x,y)] = L - M$$

3. **Constant Multiple Rule:**

$$\lim_{(x,y)\to(x_0,y_0)} kf(x,y) = kL \quad , \text{(any number } k)$$

4. **Product Rule:**

$$\lim_{(x,y)\to(x_0,y_0)} [f(x,y) \cdot g(x,y)] = L \cdot M$$

5. **Quotient Rule:**

$$\lim_{(X,y)\to(x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M} \quad , \quad M \neq 0$$

6. **Power Rule:**

$$\lim_{(x,y)\to(x_0,y_0)} [f(x,y)]^n = L^n \quad , n \text{ a positive integer}$$

7. Root Rule:

$$\lim_{(x,y)\to(x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n}$$, n a positive integer, and if n is even we assume that $L > 0$.

Continuity:

A function $f(x,y)$ is continuous at a point (x_0, y_0) if

1) $f(x,y)$ is defined at (x_0, y_0).
2) $\lim_{(x,y)\to(x_0,y_0)} f(x,y)$ exist.
3) $\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0, y_0)$

→ A function is continuous if it is continuous at every point of its domain.

Q # 20:

$$\lim_{(x,y)\to(4,3)} \frac{\sqrt{x} - \sqrt{y+1}}{x - y - 1} \quad , \quad x \neq y+1$$

$$\lim_{(x,y)\to(4,3)} \frac{\sqrt{x} - \sqrt{y+1}}{x - (y+1)}$$

$$\lim_{(x,y)\to(4,3)} \frac{\sqrt{x} - \sqrt{y+1}}{(\sqrt{x})^2 - (\sqrt{y+1})^2}$$

$$\lim_{(x,y)\to(4,3)} \frac{\cancel{\sqrt{x} - \sqrt{y+1}}}{\cancel{(\sqrt{x} - \sqrt{y+1})}(\sqrt{x} + \sqrt{y+1})}$$

$$\lim_{(x,y)\to(4,3)} \frac{1}{\sqrt{x} + \sqrt{y+1}} = \frac{1}{\sqrt{4} + \sqrt{3+1}} = \frac{1}{4}$$

Limit exists.

Two Path Method (Test) for Non-existance of Limit:

If a function $f(x,y)$ has different Limits along two different paths in the domain of $f(x,y)$ as (x,y) approaches (x_0,y_0) then

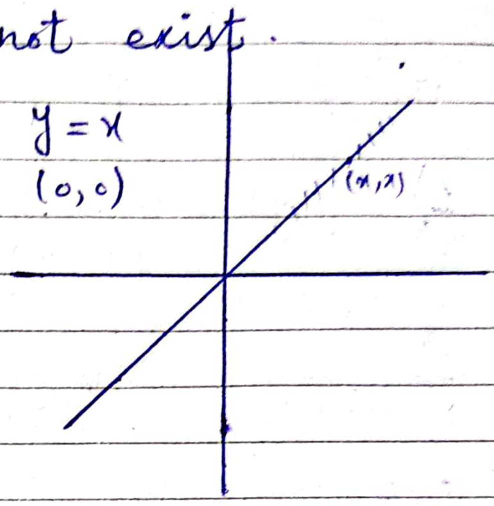
$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) \text{ does not exist.}$$

Example:

$$\lim_{(x,y)\to(0,0)} \frac{2x^2y}{x^4+y^2}$$

$y=x$

$(0,0)$



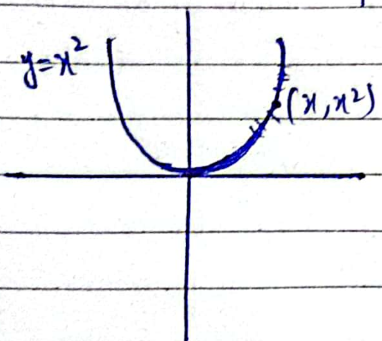
$y=0$

$$\lim_{\substack{x\to 0\\ y=0}} \frac{0}{x^4} = 0$$

$y=x$

$$\lim_{\substack{x\to 0\\ y=x}} \frac{2x^3}{x^4+x^2}$$

$$\frac{2x}{x^2+1} = 0$$



$$\lim_{\substack{x\to 0\\ y=x^2}} \frac{2x^4}{2x^4} = 1$$

Limit does not exist.

II) Method:

$y=kx^2 \quad (0,0)$

$$\lim_{\substack{x\to 0\\ y=kx^2}} \frac{2kx^4}{x^4+k^2x^4}$$

$$\lim_{\substack{x\to 0\\ y=kx^2}} \frac{2k}{1+k^2} \quad \text{Limit does not exist.}$$

Transformation:

$$x = r\cos\theta$$
$$y = r\sin\theta$$

Inverse Transformation:

$$\theta = \tan^{-1}\frac{y}{x}$$
$$r = \sqrt{x^2+y^2}$$

$$\lim_{(x,y)\to(0,0)} f(x,y) \to L$$

$$|f(x,y) - L| < \epsilon$$

$$\Rightarrow \quad 0 < \sqrt{x^2+y^2} < \delta$$

$$|f(r,\theta) - L| < \epsilon$$

$$\Rightarrow \quad 0 < |r| < \delta$$